

The point of no return

Name	Class	Date
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Watch only Everybody stays behind the line, seated, for the whole demonstration. Melted sugar is hotter than boiling water and it sticks to skin. Nobody comes forward for any reason.

Part 1 • The sugar escalation

One substance. More and more heat. Record what you SEE, HEAR and SMELL at each stage.

Stage	What is happening	What I observed	Still sugar?
1	Cold sugar		Y / N
2	Edges go clear		Y / N
3	All liquid		Y / N
4	First colour		Y / N
5	Amber		Y / N
6	Dark		Y / N

Mark the stage where you believe SUGAR STOPPED EXISTING: Stage _____

The evidence that convinced me (more than one thing):

Part 2 • The two samples

A spoonful came out at stage 3 and another at stage 5. Both have now cooled.

The STAGE 3 sample, once cool	
The STAGE 5 sample, once cool	
Which one is still sugar?	

The sugar went liquid at stage 3 AND at stage 5. Why is only one of those reversible?

If you kept heating past stage 6, would you ever get sugar back? How do you know?

YEAR 6 SCIENCE • LESSON 4

Push me Argue that the sugar transition happened one stage EARLIER or LATER than the class agreed — and defend it with evidence. What would have settled the argument?

Part 3 • Two thresholds

Two different substances this time — and we measure the temperature.

Change	Temperature	Did it come back?
Chocolate melted		
Egg white set		
Water boils	100 °C	Yes — steam condenses back to water

1. Which of these needed the LEAST heat?

2. Which one is permanent?

3. The egg set at a LOWER temperature than boiling water. So why can you un-boil water but not un-cook an egg?

4. Does knowing the temperature tell you whether a change is reversible? Explain.

The line to remember An egg cannot be un-cooked at a temperature where water is still perfectly happy being liquid.

Part 4 • Amend the chart

Should “it got hot” stay on our list as evidence that something new was made? What would you replace it with?

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1 • The sugar

Tick every box that matches what you see at each stage.

Stage	Went clear	Went runny	Changed colour	Smelled different	Made smoke
1 • Cold sugar					
2 • Edges go clear					
3 • All liquid					
4 • First colour					
5 • Amber					
6 • Dark					

Sugar stopped being sugar at stage: _____

I know because _____
_____.

2 • The temperatures

Change	Temperature	Came back?
Chocolate melted		Y / N
Egg white set		Y / N
Water boils	100 °C	YES

3 • Finish these sentences

The chocolate melted at _____ °C and it _____.

The egg white set at _____ °C and it _____.

The egg needed LESS heat than boiling water, but it is _____
_____.

This tells me that temperature on its own _____
_____.

Word bank melted • set • came back • did not come back • permanent • reversible • irreversible • new substance • does not tell you enough